

TraceMail AI

Email Forensic Investigation Report

Case ID: TM-19B34E2B
Report Generated: 2026-09-06T08:08:39.984308

1. Executive Summary

Severity	CRITICAL
Risk Score	100/100
Threat Type	Business Email Compromise / Financial Phishing

Assessment: The email presents a high-confidence set of suspicious indicators and should be treated as a potentially malicious communication. Further investigation and containment are recommended.

2. Email Evidence

Filename	suspicious_invoice.eml
Sender	finance@paypa1-security.com
Recipient	employee@company.com
Reply-To	urgent-payment@protonmail.com
Subject	URGENT: Invoice Payment Required Today
Date	Sat, 05 Sep 2026 10:30:00 +0530
Message ID	<123456789@example.com>

3. Email Authentication

Authentication Check	Result
SPF	fail
DKIM Signature	Missing
Authentication Summary	mail.company.com; spf=fail; dkim=fail; dmarc=fail

These checks help determine whether the sender's domain was properly authenticated. Failed checks increase the suspicion level but should be considered together with the other evidence.

4. Email Header & Relay Analysis

Total Relay Hops: 1
Earliest Observed Public IP: 8.8.8.8
Observed Public IP Addresses

- 8.8.8.8

Note: The earliest observed IP represents source infrastructure visible in the email headers. It should not automatically be treated as the attacker's physical location.

5. IP Intelligence

IP Address	Country	City	Network / ISP	Risk
8.8.8.8	United States	San Jose	Google LLC	LOW

IP location data describes the network infrastructure associated with the observed address. It does not necessarily identify the attacker's real-world location.

6. URL & Domain Intelligence

URLs Found: 1

- <http://paypa1-security.com/invoice/84721>

Domains Found: paypa1-security.com

The domain and URL analysis checks for suspicious naming patterns, unsafe links and indicators that may suggest impersonation or phishing activity.

7. Threat DNA

Threat DNA is a compact summary of the email's important security characteristics. It helps investigators quickly compare this email with other suspicious emails.

Security Feature	Finding
SPF Authentication	FAIL
DKIM Authentication	FAIL
DMARC Authentication	FAIL
Contains URL	Yes
Number of URLs	1
Financial / Payment Intent	Yes
Urgency Language	Yes
Reply-To Different From Sender	Yes
Contains Attachment	No
Email Relay Hops	1
Suspicious / Impersonating Domain	Yes

Threat DNA Fingerprint

SPF-FAIL|DKIM-FAIL|DMARC-FAIL|URL-1|FIN-1|URG-1|RPLY-1|ATT-0|HOP-1|TYPO-1

Detected Pattern: Business Email Compromise / Financial Phishing

8. Identified Risk Factors

The following indicators contributed to the threat assessment.

SPF authentication failed [HIGH]

DKIM authentication is missing or failed [MEDIUM]

DMARC authentication failed [HIGH]

Reply-To domain differs from sender domain [HIGH]

Urgency language detected [MEDIUM]

Detected terms: urgent, immediately, action required, today

Financial/payment-related intent detected [HIGH]

Detected terms: payment, invoice, account

1 URL(s) detected in email [MEDIUM]

9. Evidence Integrity

Integrity Method: SHA-256

Evidence Hash

59f1dc7bcd5926080987f1d49f631903072c96b47ff0b1fb6b92c4ae3c40ee61

This SHA-256 hash acts as a digital fingerprint of the analyzed email evidence. If the original evidence changes, its hash will also change, helping investigators detect unexpected modification.

10. Investigator Conclusion

The email presents a high-confidence set of suspicious indicators and should be treated as a potentially malicious communication. Further investigation and containment are recommended.

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Email Threat Detection & Forensic Intelligence Platform